Civil Engineering

Trigonometry and Complex Numbers

Assignment 1 - Questions Only

Instructions: Answer all questions. Show formula, substitution, calculation, units and interpretation.

Section A: Formula and Concept Questions

1. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$\sin \theta = \text{opposite}/\text{hypotenuse}$

2. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$\cos \theta = \text{adjacent}/\text{hypotenuse}$

3. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$\tan \theta = \text{opposite}/\text{adjacent}$

4. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$a/\sin A = b/\sin B$

5. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$c^2 = a^2 + b^2 - 2ab \cos C$

6. Explain the meaning of the formula below and identify its main symbols. (4 marks)

$z = a + bi$

Section B: Applied Questions

Problem Set 1

7. Solve a civil engineering problem involving height determination. Show formula, substitution, correct units and a technical interpretation. (10 marks)

8. Solve a civil engineering problem involving triangulation. Show formula, substitution, correct units and a technical interpretation. (10 marks)

9. Solve a civil engineering problem involving setting out angles. Show formula, substitution, correct units and a technical interpretation. (10 marks)

10. Solve a civil engineering problem involving bearings. Show formula, substitution, correct units and a technical interpretation. (10 marks)

11. Solve a civil engineering problem involving engineering phasor interpretation. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 2

12. Solve a civil engineering problem involving height determination. Show formula, substitution, correct units and a technical interpretation. (10 marks)

13. Solve a civil engineering problem involving triangulation. Show formula, substitution, correct units and a technical interpretation. (10 marks)

14. Solve a civil engineering problem involving setting out angles. Show formula, substitution, correct units and a technical interpretation. (10 marks)

15. Solve a civil engineering problem involving bearings. Show formula, substitution, correct units and a technical interpretation. (10 marks)

16. Solve a civil engineering problem involving engineering phasor interpretation. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 3

17. Solve a civil engineering problem involving height determination. Show formula, substitution, correct units and a technical interpretation. (10 marks)

18. Solve a civil engineering problem involving triangulation. Show formula, substitution, correct units and a technical interpretation. (10 marks)

19. Solve a civil engineering problem involving setting out angles. Show formula, substitution, correct units and a technical interpretation. (10 marks)

20. Solve a civil engineering problem involving bearings. Show formula, substitution, correct units and a technical interpretation. (10 marks)

21. Solve a civil engineering problem involving engineering phasor interpretation. Show formula, substitution, correct units and a technical interpretation. (10 marks)

Problem Set 4

22. Solve a civil engineering problem involving height determination. Show formula, substitution, correct units and a technical interpretation. (10 marks)

23. Solve a civil engineering problem involving triangulation. Show formula, substitution, correct units and a technical interpretation. (10 marks)

24. Solve a civil engineering problem involving setting out angles. Show formula, substitution, correct units and a technical interpretation. (10 marks)

25. Solve a civil engineering problem involving bearings. Show formula, substitution, correct units and a technical interpretation. (10 marks)

26. Solve a civil engineering problem involving engineering phasor interpretation. Show formula, substitution, correct units and a technical interpretation. (10 marks)

References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.

2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.

3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.

4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.